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Class 7 Math Notes In English Chapter – 06 The Triangle and Its Properties

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A triangle is a simple closed curve made up of three-line segments.

Classification of triangles:

On the basis of sides: A triangle is said to be

➥ an isosceles triangle if two of its sides are equal;

➥ a scalene triangle if all of its three sides are of different lengths;

➥ an equilateral triangle if all of its sides are of equal lengths.

On the basis of angles: A triangle is said to be

➥ an acute triangle if each one of its angles measure less than 90o;

➥ an obtuse triangle if one of its angles measure more than 90o;

➥ a right triangle if one of its angle's measures 90°.

Angle sum property of a triangle: The sum of all three interior angles of a triangle is 180°.

In an equilateral triangle:

➥ All sides have same length.

➥ each angle has measure 60°.

In an isosceles triangle:

- ➡ two sides have the same length.
- ➡ base angles opposite to equal sides are equal.

1. The sum of any two sides of a triangle is always greater than the third side.

2. The difference of any two sides of a triangle is always less than the third side.

3. An exterior angle of a triangle is formed when a side of a triangle is produced. There are two ways of forming an exterior angle at each vertex of a triangle.

4. Exterior angle property of a triangle: The measure of any exterior angle of a triangle is equal to the sum of the measures of its interior opposite angles.

5. In a right-angled triangle, the side opposite to the right angle is called the hypotenuse and the other two sides are called its legs.

6. Pythagoras property: In a right-angled triangle, the square of the hypotenuse is equal to the sum of the squares of the remaining two sides.

7. A median of a triangle is the line segment joining a vertex of the triangle to the mid-point of its opposite side. A triangle has three medians.

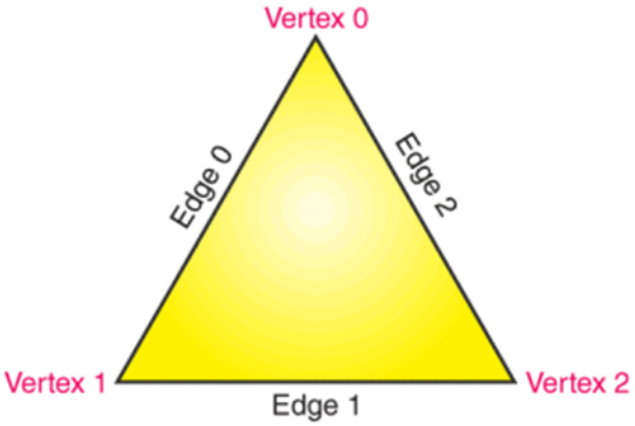
8. An altitude of triangle is the perpendicular drawn from a vertex of a triangle to its opposite sides. A triangle has three altitudes.

Triangle

A triangle is a closed curve made of three-line segments.

In Geometry, a triangle is a three-sided polygon that consists of three edges and three vertices. The most important property of a triangle is that the sum of the internal angles of a triangle is equal to 180 degrees. This property is called angle sum property of triangle.

If ABC is a triangle, then it is denoted as ΔABC, where A, B and C are the vertices of the triangle. A triangle is a two-dimensional shape, in Euclidean geometry, which is seen as three non-collinear points in a unique plane.



Shape of Triangle

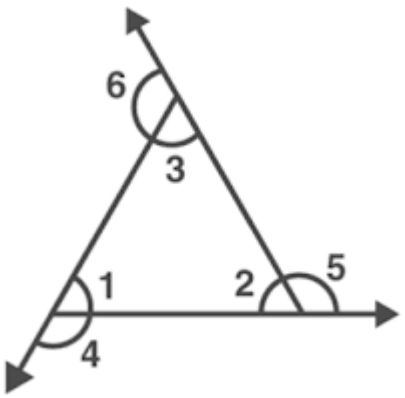
Triangle is a closed two-dimensional shape. It is a three-sided polygon. All sides are made of straight lines. The point where two straight lines join is the vertex. Hence, the triangle has three vertices. Each vertex forms an angle.

Angles of Triangle

There are three angles in a triangle. These angles are formed by two sides of the triangle, which meets at a common point, known as the vertex. The sum of all three interior angles is equal to 180 degrees.

If we extend the side length outwards, then it forms an exterior angle. The sum of consecutive interior and exterior angles of a triangle is supplementary.

Let us say, ∠1, ∠2 and ∠3 are the interior angles of a triangle. When we extend the sides of the triangle in the outward direction, then the three exterior angles formed are ∠4, ∠5 and ∠6, which are consecutive to ∠1, ∠2 and ∠3, respectively.



Hence,

$\angle 1 + \angle 4 = 180°$ ……(i)

$\angle 2 + \angle 5 = 180°$ …..(ii)

$\angle 3 + \angle 6 = 180°$ …..(iii)

If we add the above three equations, we get;

$\angle 1 + \angle 2 + \angle 3 + \angle 4 + \angle 5 + \angle 6 = 180° + 180° + 180°$

Now, by angle sum property we know,

$\angle 1 + \angle 2 + \angle 3 = 180°$

Therefore,

$180 + \angle 4 + \angle 5 + \angle 6 = 180° + 180° + 180°$

$\angle 4 + \angle 5 + \angle 6 = 360°$

This proves that the sum of the exterior angles of a triangle is equal to 360 degrees.

Properties

Each and every shape in Maths has some properties which distinguish them from each other. Let us discuss here some of the properties of triangles.

- ➥ A triangle has three sides and three angles.
- ➥ The sum of the angles of a triangle is always 180 degrees.
- ➥ The exterior angles of a triangle always add up to 360 degrees.
- ➥ The sum of consecutive interior and exterior angle is supplementary.
- ➥ The sum of the lengths of any two sides of a triangle is greater than the length of the third side. Similarly, the difference between the lengths of any two sides of a triangle is less than the length of the third side.
- ➥ The shortest side is always opposite the smallest interior angle. Similarly, the longest side is always opposite the largest interior angle.

Triangle

It has three: Sides:

Types of Triangles

On the basis of length of the sides, triangles are classified into three categories:

- ➡ Scalene Triangle
- ➡ Isosceles Triangle
- ➡ Equilateral Triangle

On the basis of measurement of the angles, triangles are classified into three categories:

- ➡ Acute Angle Triangle
- ➡ Right Angle Triangle
- ➡ Obtuse Angle Triangle

Scalene Triangle

A scalene triangle is a type of triangle, in which all the three sides have different side measures. Due to this, the three angles are also different from each other.

Isosceles Triangle

In an isosceles triangle, two sides have equal length. The two angles opposite to the two equal sides are also equal to each other.

Equilateral Triangle

An equilateral triangle has all three sides equal to each other. Due to this all the internal angles are of equal degrees, i.e. each of the angles is 60°

Acute Angled Triangle

An acute triangle has all of its angles less than 90°.

Right Angled Triangle

In a right triangle, one of the angles is equal to 90° or right angle.

Obtuse Angled Triangle

An obtuse triangle has any of its one angles more than 90°.

Important Lines in a Triangle

Median

Median is the line that connects a vertex of a triangle to the mid-point of the opposite side. In the given figure, AD is the median, joining the vertex A to the midpoint of BC.

Altitude

An altitude is a line segment through a vertex of the triangle and perpendicular to a line containing the opposite side.

Sides Also Have Constraints

Sum of the lengths of two sides of a triangle

The sum of the lengths of any two sides of a triangle is greater than the third side.

In the above triangle,

$9 + 11 = 20 > 14$

$11 + 14 = 25 > 9$

$9 + 14 = 23 > 11$

Difference between lengths of two sides of a triangle

The difference between lengths of any two sides is smaller than the length of the third side.

In the above triangle,

$11 - 9 = 2 < 14$

$14 - 11 = 3 < 9$

$14 - 9 = 5 < 11$

Using the concept of sum of two sides and difference of two sides, it is possible to determine the range of lengths that the third side can take.

Pythagoras Theorem

➥ The side opposite to the right angle in a right-angled triangle is called the hypotenuse.

➥ The other two sides are known as legs of the right-angled triangle.

➥ In a right-angled triangle, square of hypotenuse is equal to the sum of squares of legs.

$AC^2 = AB^2 + BC^2$

$\Rightarrow 5^2 = 4^2 + 3^2$

If a triangle holds Pythagoras property, then it is a right-angled triangle.

Properties of isosceles and equilateral triangles

Properties of Isosceles Triangle

Two sides are equal in length.

Base angles opposite to the equal sides are equal.

Properties of Equilateral Triangle

All three sides are equal in length.

Each angle equals to 60^0.

Classification of Triangles

Classification of triangles based on sides

Equilateral triangle: A triangle in which all the three sides are of equal lengths. Isosceles triangle: A triangle in which two sides are of equal lengths. Scalene Triangle: A triangle in which all three sides are of different length.

Classification of triangles based on angles

Acute-angled: A triangle with three acute angles.

Right-angled: A triangle with one right angle.

Obtuse-angled: A triangle with one obtuse angle.

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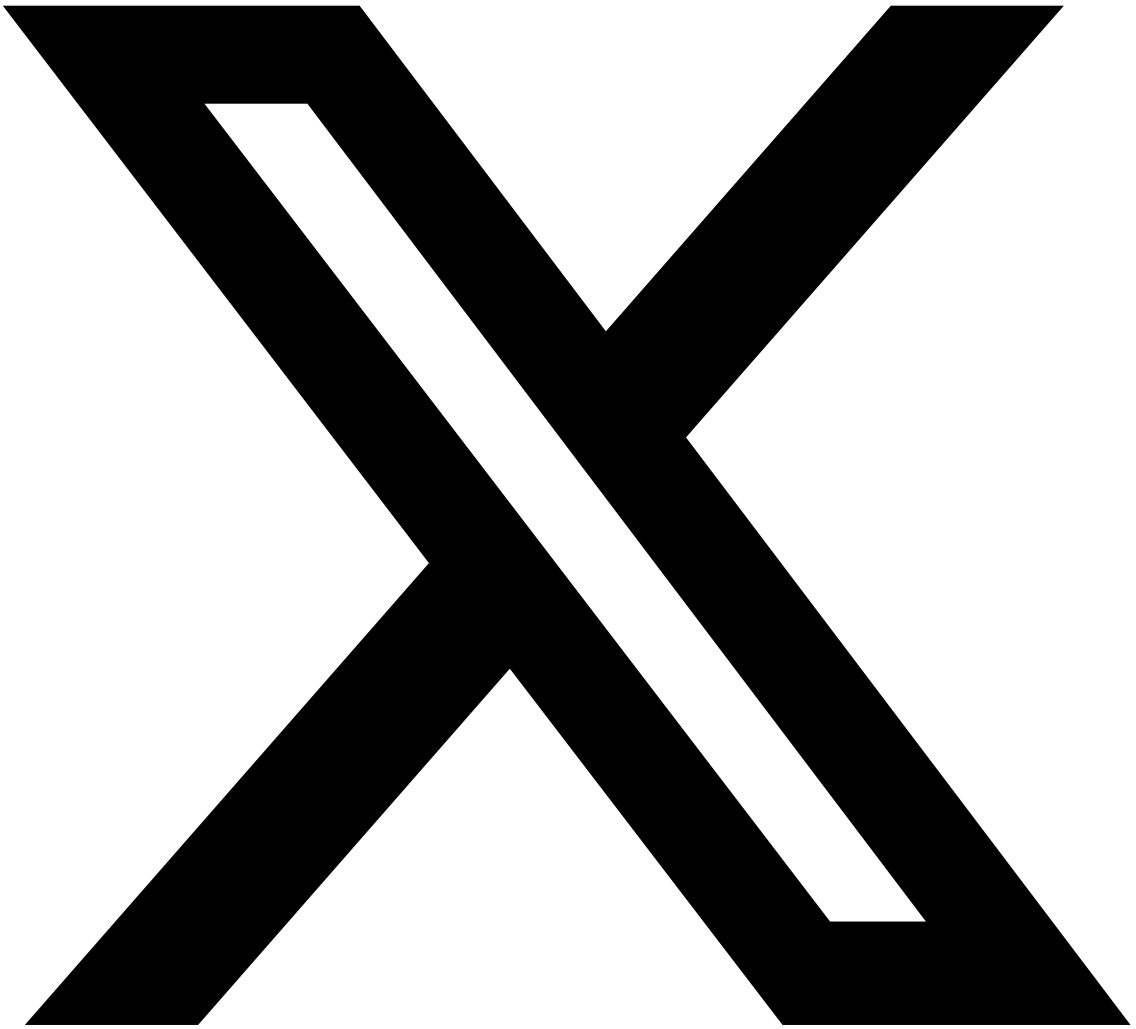

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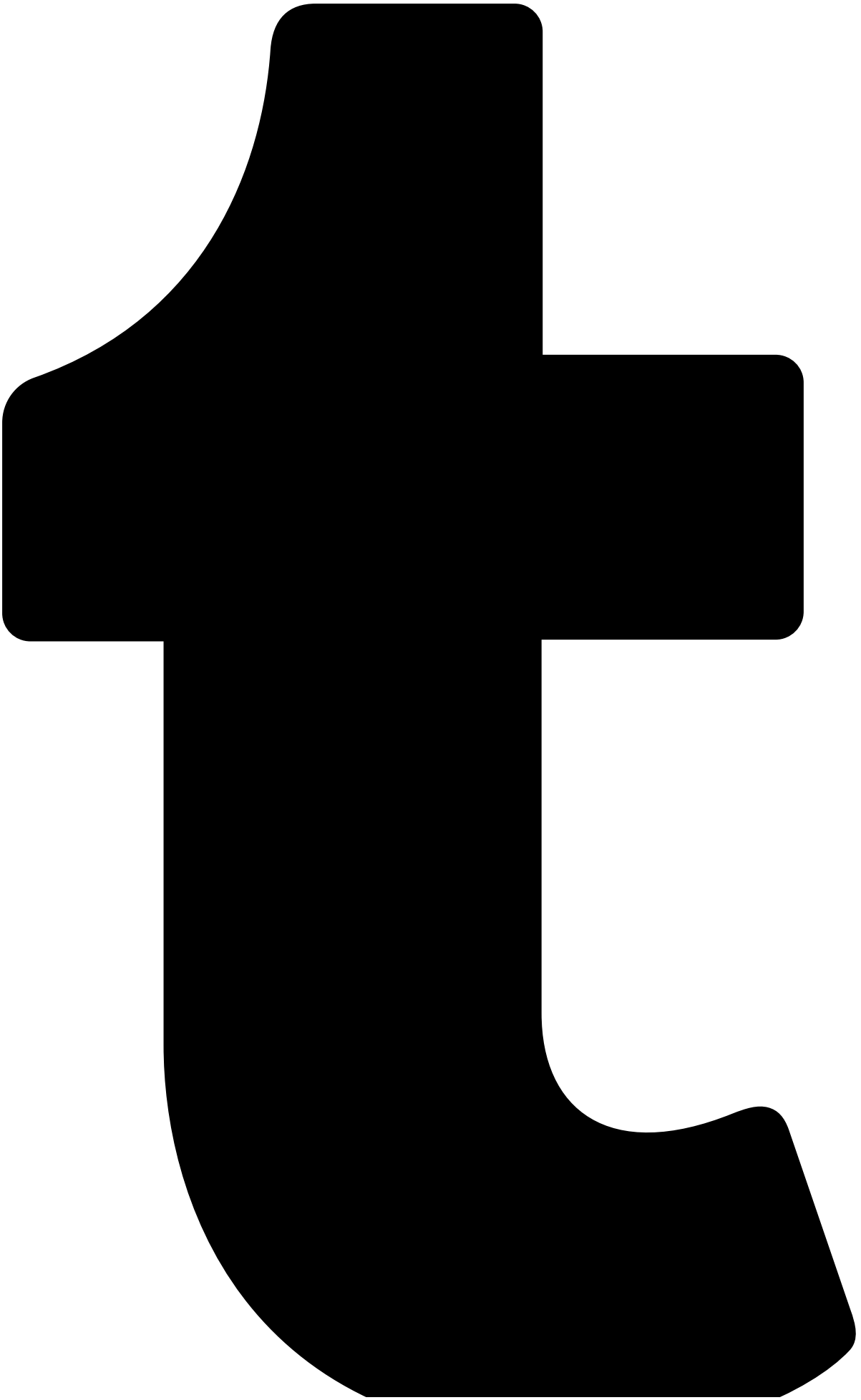

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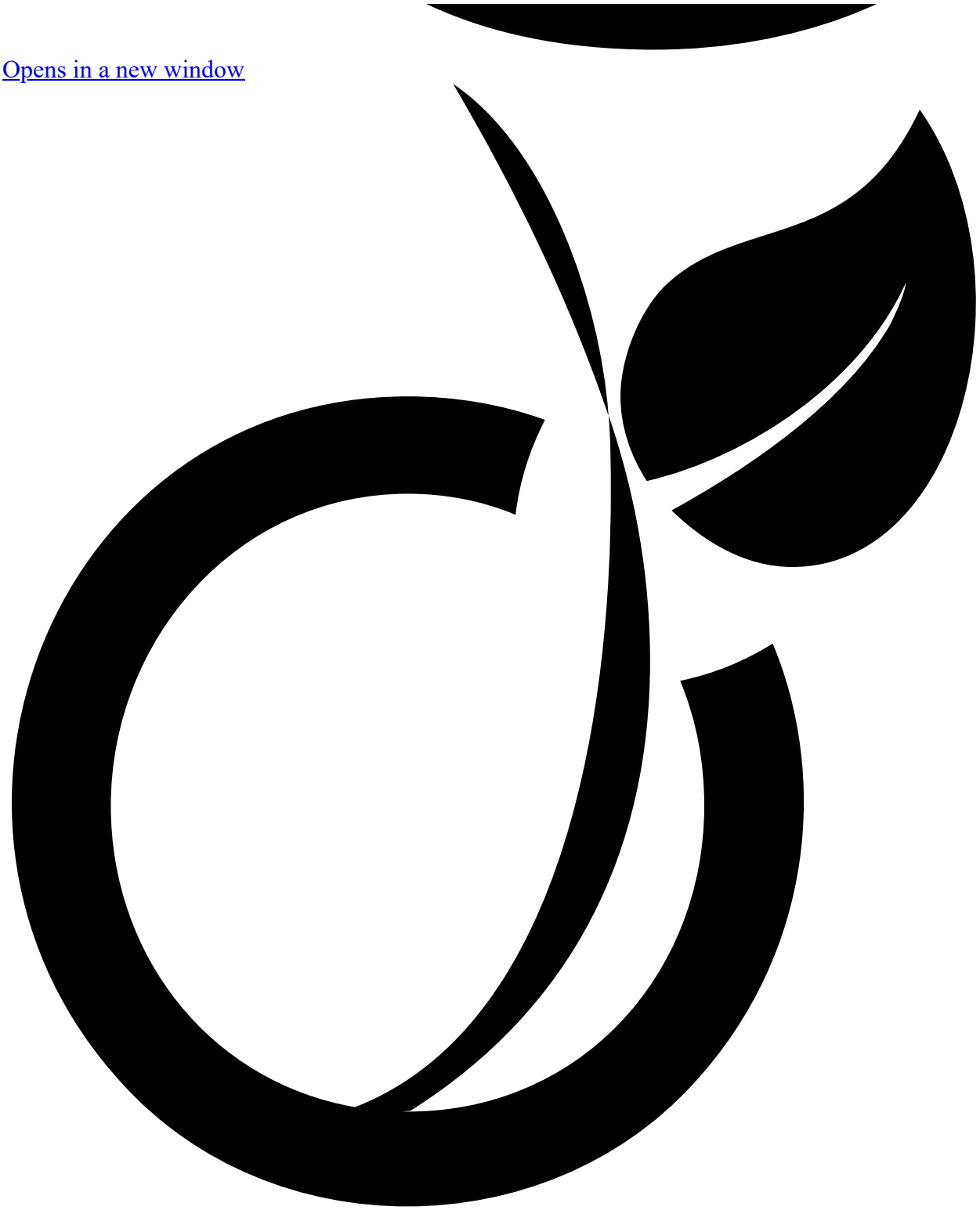

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